

1 Introduction to robotics

Although robotics is a vast field, broadly, robots are of three types – operator controlled, autonomous, and semi-autonomous.

1.1 Operator controlled

These robots are remotely controlled by a human. They do not require any sensor or processing unit as the human performs these functions and accordingly steers the robot.

1.2 Autonomous

These robots do not need any operator. They have sensors and processors in addition to actuators (motors). They sense the state around the robot, process the information and act accordingly. This category is, undoubtedly, the most fascinating category of robots as autonomous robots resemble living organisms and watching them move and doing things on their own, is a pleasure for hobbyists and robotics enthusiasts.

1.3 Semi-autonomous

These are a hybrid of autonomous and operator-controlled robots. They may or may not need sensors. Usually, they have a processor, which works in sync with the operator, and instructs the actuators accordingly. The process of making a robot requires a process-driven approach.

You need to have a complete plan of action before you begin building your robot. First, you need to study about various aspects of robot building. Next, you need to know the various methods of building the body. After that, you need to be familiar with the logic you would be implementing, and also designing the circuit. This is called ‘brainstorming’. After brainstorming, explore available resources and the best way to utilise them. The next step is designing. This involves designing the code, body and the circuit for your robot. Finally, comes the building and assembly stages of the project. If you missed out somewhere, you need to troubleshoot. The whole process would become tedious if you don’t follow a sequential approach. So, it’s always better to be systematic and follow the procedure we’ve mentioned here, especially if you’re building a robot for the first time (which we assume to be the case). To sum it up, read this book from the first page to the last, rather than jumping straight to the end. 